Lecture 1

Stochastic Analysis on Hilbert Spaces

1.1. Gaussian measures on Hilbert spaces

Preface We will use some basic facts of linear functional analysis, mostly concerning Hilbert spaces and linear operators on Hilbert spaces. These facts can be found in any introductory text book on functional analysis and essentially do not require any further understanding beyond Euclidean vector spaces, that are in fact finite-dimensional Hilbert spaces, and linear mappings on Euclidean vector spaces. The only essential difference is that in infinite dimensions linearity and continuity of linear mappings are no longer equivalent. In fact, linear mappings $L : U \to V$ between Hilbert spaces U and V (or any normed vector spaces) are continuous if and only if they are bounded, i.e. $L(B_1(0))$ is contained in a bounded subset or equivalently,

$$\sup_{\|h\|_U \leq 1} \|Lh\|_V < \infty\,.$$

Definition 1.1. $(H, \langle \cdot, \cdot \rangle_H)$ is called a **(real) Hilbert space**, if:

(i) H is a vector space over $\mathbb{R}$,

(ii) $\langle \cdot, \cdot \rangle_H : H \times H \longrightarrow \mathbb{R}$ is a scalar product, i.e., a symmetric positive definite bilinear form. In particular, $|u|_H := \sqrt{\langle u, u \rangle_H}$ defines a norm on H.

(iii) $(H, |\cdot|_H)$ is a complete normed (real) vector space, hence a (real) Banach space.

Example 1.2. (i) $\mathbb{R}^N$, equipped with usual scalar product $\langle u, v \rangle := \sum\limits_{i=1}^{N} u_i v_i$, but also with $\langle u, v \rangle_Q := \sum\limits_{i,j=1}^{N} q_{ij} u_i v_j$, where $Q = (q_{ij})$ is any symmetric postivie definite $N \times N$-matrix.

(ii) Let $(\Omega, \mathcal{A}, \mu)$ be a measure space, then $L^2(\Omega, \mathcal{A}, \mu) =: L^2(\mu)$ with scalar product $\langle f, g \rangle := \int\limits_{\Omega} fg d\mu$ is a real Hilbert space, but also with the scalar product

$$\langle f, g \rangle_Q := \int\limits_{\Omega} \int\limits_{\Omega} f(x) g(y) q(x, y) \mu(dx) \mu(dy)$$

where q is symmetric and positive definite in the sense that $\forall\, A_1, \ldots, A_N \in \mathcal{A}$ the matrix $Q = (q_{ij})$, with elements $q_{ij} = \int_{A_i} \int_{A_j} q(x, y)\, \mu(dx) \mu(dy) \geq 0$, $1 \leq i, j \leq N$, is symmetric and positive definite.

(iii) The sequence space $l^2 = \left\{ (u_k)_{k \geq 1} \subset \mathbb{R} : \sum\limits_{k=1}^{\infty} u_k^2 < \infty \right\}$ equipped with

$$\langle u, v \rangle := \sum_{k=1}^{\infty} u_k v_k.$$

Definition 1.3. $(H, \langle \cdot, \cdot \rangle_H)$ is called **separable**, if there exists a countable dense subset $H_0 \subset H$.

EXAMPLE 1.4. $L^2([0,1],dx)$ is separable, with countable dense subsets

$$H_0 = \left\{ \sum_{k=0}^{n} a_k x^k : n \geq 1, a_k \in \mathbb{Q} \right\} \quad = \text{all polynomials with rational coefficients}$$

or

$$H_0 = \left\{ \sum_{k=1}^{n} a_k \sin(2\pi kx) + \sum_{k=0}^{n} b_k \cos(2\pi kx) : n \geq 1, a_k, b_k \in \mathbb{Q} \right\} \quad \begin{array}{c}= \text{all Fourier polynomials with} \\ \text{rational coefficients}\end{array}$$

In general: If Ω is a separable metric space and $\mathcal{A} = \mathcal{B}(\Omega)$ the Borel σ-algebra, then $L^2(\mu)$ is separable.

DEFINITION 1.5. A probability measure μ on $(H, \mathcal{B}(H))$ is called a **Gaussian measure**, if for all $h \in H$ the linear functional

$$l_h : g \mapsto \langle g, h \rangle_H, \quad H \longrightarrow \mathbb{R}$$

is normally distributed, i.e., there exist $m(h) \in \mathbb{R}$, $\sigma^2(h) \in \mathbb{R}_+$, such that for all $t \in \mathbb{R}$:

$$\int_H e^{itl_h} d\mu = e^{itm(h) - \frac{1}{2}t^2\sigma^2(h)}.$$

THEOREM 1.6. *A probability measure μ on $(H, \mathcal{B}(H))$ is a Gaussian measure if and only if*

$$\int_H e^{i\langle g,h\rangle_H} \mu(dg) = e^{i\langle m,h\rangle_H - \frac{1}{2}\langle Qh,h\rangle_H}$$

for all $h \in H$, where

a) *$m \in H$ is the mean vector,*
b) *$Q : H \longrightarrow H$ is linear, continuous and symmetric (i.e. $\langle Qg, h\rangle_H = \langle g, Qh\rangle_H$), positive semidefinite (i.e. $\langle Qg, g\rangle_H \geq 0 \ \forall g \in H$) and of finite trace, i.e.*

$$tr(Q) = \sum_{k=1}^{\infty} \langle Qe_k, e_k\rangle_H < \infty$$

for one (and hence for all) complete orthonormal system (CONS) (e_k) on H.

In this case we will write

$$\mu = \mathcal{N}(m, Q)$$

and Q is called the covariance operator of μ.

In particular:

(i) $\int_H \langle x, h\rangle_H \mu(dx) = \mathbb{E}_\mu(l_h) = m(h) = \langle m, h\rangle_H$

(ii)

$$\begin{aligned}\int_H (\langle x, h_1\rangle - \langle m, h_1\rangle)(\langle x, h_2\rangle - \langle m, h_2\rangle)\mu(dx) &= \mathbb{E}_\mu\left((l_{h_1} - \mathbb{E}_\mu l_{h_1})(l_{h_2} - \mathbb{E}_\mu l_{h_2})\right) \\ &= Cov_\mu(l_1, l_2) \\ &= \langle Qh_1, h_2\rangle_H \quad \forall h_1, h_2 \in H\end{aligned}$$

(iii) $\int_H \|x - m\|_H^2 \mu(dx) = tr(Q)$

Example 1.7. (i) **Wiener-measure**

Let $(\beta(t))_{t\geq 0}$ be a 1-dimensional continuous **Brownian motion** on $(\Omega, \mathcal{F}, \mathbb{P})$, i.e. a stochastic process satisfying

a) $\beta(0) = 0$ $\mathbb{P}$-a.s.,
b) for $0 \leq t_1 < t_2 < ... < t_n$ the increments

$$\beta(t_{i+1}) - \beta(t_i), \qquad 1 \leq i \leq n-1$$

are independent and $\mathcal{N}(0, t_{i+1} - t_i)$ distributed.
c) $t \mapsto \beta(t)$ is $\mathbb{P}$-a.s. continuous.

Let $T > 0$ be fixed and let

$$\beta_\cdot : \Omega \longrightarrow C([0,T]) \subseteq L^2([0,T]),\ \omega \mapsto (t \mapsto \beta(t,\omega))$$

be measurable w.r.t. the Borel σ-algebra on $L^2([0,T])$. Then the distribution

$$\mu(A) := \mathbb{P}(\beta(\cdot) \in A), \qquad A \in \mathcal{B}(L^2([0,T]))$$

is called the **Wiener measure**.

μ is a Gaussian measure (centered, i.e., the mean vector is zero) with covariance operator Q induced by the bilinear form

$$\begin{aligned}\langle Qg, h\rangle_{L^2([0,T])} &= \int_0^T Qg(t)h(t)\,dt \\ &= \int_0^T \int_0^T g(s)h(t)\, s \wedge t\, dsdt \\ &= \langle (-\Delta)^{-1} g, h\rangle_{L^2([0,T])}\end{aligned}$$

where Δ denotes the Laplace-Operator on $L^2([0,T])$ (i.e. $\Delta g = g''$), with Dirichlet boundary conditions in 0 ($g(0) = 0$) and Neumann boundary conditions in T ($g'(T) = 0$). With these boundary conditions, Δ becomes invertible and we have indeed that $Q = (-\Delta)^{-1}$.

Proof.

$$\begin{aligned}\mathbb{E}_\mu(l_h) &= \int_\Omega l_h(\beta) d\mathbb{P} = \int_\Omega \langle h, \beta\rangle_{L^2([0,T])} d\mathbb{P} \\ &= \int_\Omega \int_0^T h(s)\beta(s) ds d\mathbb{P} = \int_0^T h(s) \int_\Omega \beta(s) d\mathbb{P} ds \\ &= \int_0^T h(s) \underbrace{\mathbb{E}_\mu(\beta(s))}_{=0} ds = 0, \text{ hence } m = 0.\end{aligned}$$

$$\mathrm{Cov}_\mu(l_g, l_h) = \underbrace{\int\limits_{L^2([0,T])} \langle x, g\rangle\langle x, h\rangle \mu(dx)}_{\text{since } m(g)=m(h)=0} = \mathbb{E}\left(\int\limits_0^T g(s)\beta(s)ds \int\limits_0^T h(s)\beta(s)ds\right)$$

$$= \int\limits_0^T\int\limits_0^T g(s)h(t)\,\underbrace{\mathbb{E}(\beta(s)\beta(t))}_{=s\wedge t}\,dsdt.$$

□

Remark It is instructive to identify Q from the bilinear form. Indeed,

$$\langle Qg, h\rangle_{L^2([0,T])} = \int\limits_0^T\int\limits_0^T g(s)h(t)\, s\wedge t\, dsdt = \int\limits_0^T \left(\int\limits_0^T g(s)\, s\wedge t\, ds\right) h(t)\, dt$$

implies that

$$Qg(t) = \int_0^T g(s)s\wedge t\, ds = \int_0^t g(s)s\, ds + t\int_t^T g(s)\, ds.$$

In particular, $Qg(0) = 0$, $(Qg)'(t) = \int_t^T g(s)\, ds$, which is equal to 0 if $t = T$, and $(Qg)''(t) = -g(t)$.

(ii) **Brownian bridge**

The process

$$\beta^0(t) := \beta(t) - \frac{t}{T}\beta(T), \quad 0 \le t \le T$$

satisfies $\beta^0(T) = 0$ and is therefore called a **Brownian bridge** from 0 to 0 (in time T). The distribution

$$\mu^0(A) := \mathbb{P}(\beta^0(\cdot) \in A), \quad A \in \mathcal{B}(L^2([0,T]))$$

is called a **pinned Wiener measure**.

μ^0 is a Gaussian measure with mean vector 0 and covariance operator

$$\mathrm{Cov}_{\mu^0}(l_g, l_h) = \langle Q^0 g, h\rangle = \int\limits_0^T\int\limits_0^T g(s)h(t)\left(\ s\wedge t\ - \frac{st}{T}\right) dsdt$$
$$= \langle(-\Delta_D)^{-1}g, h\rangle$$

where Δ_D now denotes the Dirichlet Laplacian on $L^2([0,T])$. In particular, $Q^0 = (-\Delta_D)^{-1}$.

Proof.

$$\mathrm{Cov}_{\mu^0}(l_g, l_h) = \mathbb{E}_{\mu^0}(l_g, l_h) = \int\limits_0^T\int\limits_0^T g(s)h(t)\mathbb{E}(\beta^0(s)\beta^0(t))dsdt$$

with

$$\begin{aligned}\mathbb{E}(\beta^0(s)\beta^0(t)) &= \mathbb{E}\Big((\beta(s)-\frac{s}{T}\beta(T))(\beta(t)-\frac{t}{T}\beta(T))\Big)\\ &= \mathbb{E}(\beta(s)\beta(t)) - \frac{t}{T}\mathbb{E}(\beta(s)\beta(T)) - \frac{s}{T}\mathbb{E}(\beta(T)\beta(t)) + \frac{st}{T^2}\mathbb{E}(\beta^2(T))\\ &= s\wedge t \ - \frac{st}{T} - \frac{st}{T} + \frac{st}{T}\\ &= s\wedge t \ - \frac{st}{T}.\end{aligned}$$

□

In this case

$$Q^0g(t) = \int_0^T g(s)\left(s\wedge t - \frac{st}{T}\right)ds = \int_0^t g(s)s\,ds + t\int_t^T g(s)\,ds - \frac{t}{T}\int_0^T g(s)s\,ds$$

indeed satisfies $Q^0g(0) = Q^0g(T) = 0$ and $(Qg)'' = -g$.

Sketch of proof of theorem 1.6.
⇐: obvious

⇒: First we have to show that $h \mapsto m(h) = \mathbb{E}_\mu l_h$ is linear and continuous.

linear:

$$\begin{aligned}m(h+g) = \mathbb{E}_\mu(l_{h+g}) &\overset{\text{scalar prod.}}{\underset{\text{linear}}{=}} \mathbb{E}_\mu(l_h + l_g)\\ &\overset{\text{exp.}}{\underset{\text{linear}}{=}} \mathbb{E}_\mu(l_h) + \mathbb{E}_\mu(l_g)\\ &= m(h) + m(g)\end{aligned}$$

continuous:
$m(h)$ is continuous (on H) if and only if $\sup\limits_{\|h\|_H\le 1} m(h) < \infty$.
First note that:

$$H_n := \left\{h\in H : \int |l_h(x)|\mu(dx) \le n\right\} \subseteq H$$

is closed for all $n \ge 1$ according to Fatou's lemma. Moreover

$$H = \bigcup_{n\ge 1} H_n.$$

Therefore Baire's category theorem implies that there exists some $n_0 \ge 1$ such that

$$\overset{\circ}{H}_{n_0} \ne \emptyset.$$

Hence we can find $\varepsilon > 0$ and some element $h_0 \in H$ such that $B_\varepsilon(h_0) \subseteq H_{n_0}$. But this implies that for all $h \in H$ with $\|h\|_H \le 1$:

$$h = \frac{2}{\varepsilon}\left(\frac{\varepsilon}{2}h\right) = \frac{2}{\varepsilon}\Big(\underbrace{\frac{\varepsilon}{2}h + h_0}_{\in B_\varepsilon(h_0)} - h_0\Big).$$

Finally this implies that

$$\begin{aligned}|m(h)| &= \left|\int_H \langle x,h\rangle_H \mu(dx)\right| \\ &\le \frac{2}{\varepsilon}\Big(\underbrace{\int_H |l_{\frac{\varepsilon}{2}h+h_0}|\mu}_{\le n_0} + \underbrace{\int_H |l_{h_0}|d\mu}_{\le n_0}\Big) \\ &\le \frac{4}{\varepsilon} n_0 < \infty.\end{aligned}$$

Therefore $\sup_{\|h\|_H\le 1} m(h) \le \frac{4}{\varepsilon}n_0 < \infty$ which implies that $m(h)$ is continuous. The Riesz representation theorem now implies that there exists $m \in H$ such that

$$\langle m,h\rangle_H = m(h).$$

Similarly one can show that Q is bilinear and continuous on $H \times H$. In particular, (again using Riesz representation theorem):

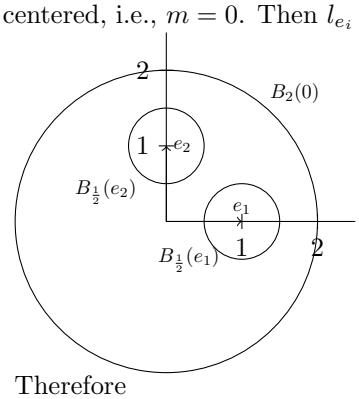
$$\forall h \in H\ \exists Q(h) : \langle Q(h), g\rangle = Q(h,g) \quad \forall g \in H$$

and therefore $h \mapsto Q(h)$ is linear and continuous.

Q is positive definite, since

$$\langle Qh,h\rangle_H = \int_H (l_h(x) - \langle x,m\rangle_H)^2 \mu(dx) \ge 0.$$

It remains to show that $\mathrm{tr}(Q) < \infty$. We will give a rigorous proof in the subsequent Lemma which is of its own interest. Here we only give a rough sketch, why the finite trace condition shows up. To this end let $(e_n)_n$ be a CONS and let μ be centered, i.e., $m = 0$. Then $l_{e_i} \sim \mathcal{N}(0, \langle Qe_i, e_i\rangle)$ and for given e_i we have that:



Therefore

$$\bigcup_{i\ge 1} B_{\frac{1}{2}}(e_i) \subseteq B_2(0).$$

Consequently, □

$$\infty > \mu(B_2(0)) \ge \sum_{i=1}^{\infty} \mu(B_{\frac{1}{2}}(e_i)) \sim \sum_{i=1}^{\infty} \langle Qe_i, e_i\rangle.$$

LEMMA 1.8. *Let $\mu = N(0,Q)$ be a Gaussian measure on H. Then*

$$1 + tr\ (Q) \leq \left(\int e^{-\frac{\|x\|^2}{2}}\ \mu(dx) \right)^{-2}. \tag{1.20}$$

In particular $tr(Q) < \infty$.

PROOF. We first assume that $\dim(H) = n < \infty$. Let $\sigma_1^2, \ldots, \sigma_n^2$ be the eigenvalues of Q. W.l.o.g. we may assume that $\sigma_i^2 > 0$. Then

$$\begin{aligned}
\int e^{-\frac{\|x\|^2}{2}}\ \mu(dx) &= \frac{1}{\sqrt{(2\pi)^n \det(Q)}} \int e^{-\sum_{i=1}^n \frac{x_i^2}{2} + \frac{x_i^2}{2\sigma_i^2}}\ dx \\
&= \frac{1}{\sqrt{(2\pi)^n \det(Q)}} \int e^{-\sum_{i=1}^n \frac{\sigma_i^2+1}{\sigma_i^2}\frac{x_i^2}{2}}\ dx \\
&= \prod_{i=1}^n \frac{1}{\sqrt{1+\sigma_i^2}}.
\end{aligned}$$

Hence

$$\prod_{i=1}^n (1+\sigma_i^2) = \left(\int e^{-\frac{\|x\|^2}{2}}\ \mu(dx) \right)^{-2}$$

and thus

$$1 + \mathrm{tr}\ (Q) = 1 + \sum_{i=1}^n \sigma_i^2 \leq \prod_{i=1}^n (1+\sigma_i^2) = \left(\int e^{-\frac{\|x\|^2}{2}}\ \mu(dx) \right)^{-2}$$

which proves the assertion in the finite dimensional case.

In the general case consider $\mu_n = N(0, Q_n)$, the distribution of μ w.r.t. the projection $\Pi_n : H \to \mathbb{R}^n$, $h \mapsto (\langle h, e_1\rangle, \ldots, \langle h, e_n\rangle)$, for a given CONS (e_k) of H. The first step now implies for all n that

$$\begin{aligned}
1 + \mathrm{tr}\ (Q_n) &\leq \left(\int_{\mathbb{R}^n} e^{-\frac{\|x\|^2}{2}}\ \mu_n(dx) \right)^{-2} \\
&= \left(\int_H e^{-\frac{1}{2}\sum_{i=1}^n \langle h, e_i\rangle^2}\ \mu(dh) \right)^{-2}.
\end{aligned}$$

Taking the limit $n \to \infty$ yields the assertion, since

$$\lim_{n\to\infty} \mathrm{tr}\ (Q_n) = \lim_{n\to\infty} \sum_{i=1}^n \langle Qe_i, e_i\rangle = \sum_{i=1}^\infty \langle Qe_i, e_i\rangle = \mathrm{tr}\ (Q)$$

and

$$\lim_{n\to\infty} \int_H e^{-\frac{1}{2}\sum_{i=1}^n \langle h, e_i\rangle^2}\ \mu(dh) = \int e^{-\frac{\|h\|^2}{2}}\ \mu(dh).$$

Note that the r.h.s. is indeed strictly positive, since $\mu(\|h\| < \infty) = 1$. This is the only place in the proof, where the support property of μ was actually used (!). $\square$

REMARK 1.9. The finite trace condition excludes in particular the existence of the standard normal distribution on an infinite dimensional Hilbert space. This can be seen also directly from the asymptotics of the volume of n-dimensional balls

w.r.t. standard normal distribution. Indeed a straightforward computation leads to

$$N(0, Id_n)(B_R(0)) = \frac{1}{\sqrt{2\pi}^n} \int_{B_R(0)} e^{-\frac{\|x\|^2}{2}}\, dx = \frac{\sigma_n}{\sqrt{2\pi}^n} \int_0^R r^{n-1} e^{-\frac{r^2}{2}}\, dr$$

where $\sigma_n = \frac{2\sqrt{\pi}^n}{\Gamma\left(\frac{n}{2}\right)}$ denotes the surface of the unit sphere $S^{n-1} = \{x \in \mathbb{R}^n : \|x\| = 1\}$ in $\mathbb{R}^n$. Consequently

$$N(0, Id_n)(B_R(0)) = \frac{1}{\Gamma\left(\frac{n}{2}\right)} \int_0^{\frac{R^2}{2}} s^{\frac{n}{2}-1} e^{-s}\, ds \to 0$$

for all finite R.

So if there would exists a standard normal distribution μ on an infinite dimensional Hilbert space H, it would follow that $\mu(B_R(0)) = 0$ for all finite R, which is a contradiction to $\mu(H) = 1$. With the same argument one could also show that there can be no Gaussian measure $\mu = N(0, Q)$ on H with bounded inverse Q^{-1}, since $\sqrt{Q}^{-1} x \sim N(0, Id_H)$ implies

$$N(0, Q)(B_R(0)) \leq \lim_{n\to\infty} N(0, Id_n)(B_{\|\sqrt{Q}^{-1}\| R}(0)) = 0$$

again for all R.

So in order to realize a Gaussian measure μ with covariance operator Id_H on an infinite dimensional Hilbert space requires to embed H into some larger Hilbert space H_1, such that the embedding $J : H \hookrightarrow H_1$ becomes Hilbert Schmidt. (For more details see below the remark on stochastic integrals w.r.t. cylindrical Wiener processes).

We will use the following functional analytic fact on trace-class operators without further proof:

Proposition 1.10. *Let Q be linear, continuous, symmetric, positive semidefinite and let $tr(Q) < \infty$ (i.e. $tr(Q) = \sum_{i=1}^{\infty} \langle Qe_i, e_i\rangle < \infty$ where (e_i) is any CONS). Then there exists a CONS e_k, $k \geq 1$, of H, consisting of eigenvectors of Q, i.e.,*

$$Qe_k = \lambda_k e_k, \quad \lambda_k \geq 0.$$

0 is the only accumulation point of the sequence of eigenvalues $(\lambda_k)_{k\geq 1}$. Moreover, $tr(Q) = \sum\limits_{i=1}^{\infty} \lambda_k < \infty$, i.e. (λ_k) converges absolutely.

The following proposition provides the canonical representation of Gaussian measures on Hilbert spaces and also gives an instruction of how to simulate Hilbert space valued Gaussian random variables, resp. to draw samples from a Gaussian measure on some Hilbert space.

Proposition 1.11 (canonical representation). *Let $m \in H$ and let Q and $(e_k)_{k\geq 1}$ be as in the previous Proposition 1.10. Let X be an H-valued random variable on $(\Omega, \mathcal{F}, \mathbb{P})$. Then X is normally distributed with $X \sim \mathcal{N}(m, Q)$ if and only if*

$$X = \sum_{k=1}^{\infty} \sqrt{\lambda_k} \beta_k e_k + m,$$

where $(\beta_k)_{k\geq 1}$ are independent, $\mathcal{N}(0, 1)$ distributed random variables. The infinite series converges $\mathbb{P}$-a.s. and in L^p for all $p < \infty$.

PROOF. $\Leftarrow$**:** We have to show that $\sum_{k=1}^{\infty} \sqrt{\lambda_k}\beta_k e_k + m$ converges on H. To this end let $m \leq n \in \mathbb{N}$. Then:

$$\begin{aligned}
\|\sum_{k=1}^{n} \sqrt{\lambda_k}\beta_k - \sum_{k=1}^{m} \sqrt{\lambda_k}\beta_k\|_H^2 &= \|\sum_{k=m+1}^{n} \sqrt{\lambda_k}\beta_k\|_H^2 \\
&= \langle \sum_{k=m+1}^{n} \sqrt{\lambda_k}\beta_k, \sum_{k=m+1}^{n} \sqrt{\lambda_k}\beta_k \rangle_H \\
&= \sum_{k,l=m+1}^{n} \sqrt{\lambda_k}\sqrt{\lambda_l}\beta_k\beta_l \underbrace{\langle e_k, e_l \rangle_H}_{=\delta_{kl}} \\
&= \sum_{k=m+1}^{n} \lambda_k \beta_k^2.
\end{aligned}$$

Hence

$$\begin{aligned}
\mathbb{E}\left(\|\sum_{k=1}^{n} \sqrt{\lambda_k}\beta_k - \sum_{k=1}^{m} \sqrt{\lambda_k}\beta_k\|_H^2 \right) &= \mathbb{E}\left(\sum_{k=m+1}^{n} \lambda_k \beta_k^2 \right) \\
&= \sum_{k=m+1}^{n} \lambda_k \underbrace{\mathbb{E}\beta_k^2}_{=1} = \sum_{k=m+1}^{n} \lambda_k \overset{n,m\to\infty}{\longrightarrow} 0
\end{aligned}$$

Finally we have to verify that the infinite series has the correct distribution:

$$\begin{aligned}
\mathbb{E}e^{i\langle h, X\rangle_H} &= \mathbb{E}e^{i\langle m,h\rangle_H + i\sum_{k=1}^{\infty}\sqrt{\lambda_k}\beta_k\langle e_k,h\rangle_H} \\
&= e^{i\langle m,h\rangle_H} \prod_{k=1}^{\infty} \underbrace{\mathbb{E}e^{i\sqrt{\lambda_l}\langle e_k,h\rangle_H\beta_k}}_{=e^{-\frac{\lambda_k}{2}\langle e_k,h\rangle_H^2}} \\
&= e^{i\langle m,h\rangle_H} e^{-\frac{1}{2}\sum_{k=1}^{\infty}\lambda_k\langle e_k,h\rangle_H^2} \\
&= e^{i\langle m,h\rangle_H} e^{-\frac{1}{2}\sum_{k=1}^{\infty}\lambda_k\langle e_k,h\rangle_H\langle e_k,h\rangle_H} \\
&= e^{i\langle m,h\rangle_H} e^{-\frac{1}{2}\sum_{k=1}^{\infty}\langle \lambda_k e_k,h\rangle_H\langle e_k,h\rangle_H} \\
&= e^{i\langle m,h\rangle_H} e^{-\frac{1}{2}\sum_{k=1}^{\infty}\langle Qe_k,h\rangle_H\langle e_k,h\rangle_H} \\
&= e^{i\langle m,h\rangle_H} e^{-\frac{1}{2}\sum_{k=1}^{\infty}\langle e_k,Qh\rangle_H\langle e_k,h\rangle_H} \\
&\underset{\text{identity}}{\overset{\text{Parsevals'}}{=}} e^{i\langle m,h\rangle_H} e^{-\frac{1}{2}\langle Qh,h\rangle_H}
\end{aligned}$$

$\square$